

Mentis Umbra: Consciousness, Agency, and Qualia as Admissible Projections

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Abstract

We establish a structural classification theorem: any admissible irreversible system that is redescription-stable must instantiate a projection structure consisting of a transition mediator and an invariant reference. We further extend the result to interacting multi-system cases, proving existence and necessity-under-removal of interaction mediators and jointly invariant references, lifting to n -system settings via pairwise restriction, and giving a constructive global mediation criterion via triple compatibility. Terminology relating to agency, intentionality, and interior representation is introduced only as role-based naming conventions for derived structural predicates. No new primitives are introduced.

1 Framework Axioms

This paper assumes the Core Framework Axioms stated in Appendix A. All references to admissibility, standing, and redescription stability are governed by these axioms.

2 Quotient-Mediator Clarification

Let $\approx$ denote equivalence under admissible redescription: $X \approx Y$ iff $\exists R$ admissible with $R(X) \sim Y$.

Define the canonical projection

$$\pi : S \rightarrow S/\approx.$$

Lemma 1 (Descent to Quotient). If redescription stability holds, then for every $T \in E(S)$ there exists a unique induced map $\hat{T}$ such that

$$\pi(T(S)) = \hat{T}(\pi(S)).$$

Proof. If $X \approx Y$, then $Y \sim R(X)$ for some admissible R . By redescription stability,

$$T(Y) \sim T(R(X)) \sim R(T(X)).$$

Hence $T(X) \approx T(Y)$, so T is constant on $\approx$ -classes and descends uniquely. $\square$

This quotient projection provides the canonical mediator structure referenced throughout the paper.

3 Single-System Framework (Fully Formal)

3.1 Typing/Applicability and Irreversibility

Definition 1 (Single-System Envelope). Let S be a system. Define $\mathcal{E}(S)$ to be the set of transition operators T such that the application $T(S)$ is well-defined and yields a system of the same structural type as S . This is a typing/applicability domain only; it imposes no semantic condition on transition outcomes.

Remark (Presentation Regime Dichotomy). Statewise envelope non-minimality is a syntactic condition on representation: it asserts absence of a canonical presentation at each well-typed state. If a canonicalization discipline were imposed (eliminating all non-identity alpha-variants), then the non-minimality condition would fail and the determinacy argument would trivialize. Thus the results bifurcate cleanly between canonical and non-canonical presentation regimes.

Definition 2 (Irreversible Transition). A transition $T \in \mathcal{E}(S)$ is *irreversible* if there does not exist any admissible transition $U \in \mathcal{E}(T(S))$ such that

$$U(T(S)) \sim S.$$

3.2 Equivalence, Redescription, and Stability

Definition 3 (Identity Equivalence). For systems S_1, S_2 , we write

$$S_1 \sim S_2$$

iff:

- (i) standing status matches,
- (ii) $\mathcal{E}(S_1) = \mathcal{E}(S_2)$,
- (iii) for every $T \in \mathcal{E}(S_1)$, the outcomes $T(S_1)$ and $T(S_2)$ agree up to alpha-equivalence of internal representation symbols only.

No semantic drift beyond alpha-equivalence is permitted.

Definition 4 (Admissible Redescription). A transformation R on system representations is *admissible* if:

- (i) standing status is preserved,
- (ii) typing/applicability envelope is preserved:

$$\mathcal{E}(S) = \mathcal{E}(R(S)),$$

- (iii) no anchors are introduced and none are removed,
- (iv) identity equivalence under $\sim$ is preserved (alpha-equivalence only).

Definition 5 (Redescription Stability). A system S is *redescription-stable* if for every admissible redescription R and every $T \in \mathcal{E}(S)$,

$$R(T(S)) \sim T(R(S)).$$

3.3 Mediator Factorization and Invariant Reference

Definition 6 (Transition Mediator). A structure P is a *mediator* for S if for every $T \in \mathcal{E}(S)$ there exists an induced operator $\hat{T}$ such that

$$T(S) = \hat{T}(P(S)).$$

Lemma 2 (Nontrivial Admissible Renaming). If evaluation of some $T \in \mathcal{E}(S)$ depends on internal representation labels beyond alpha-equivalence, then there exists an admissible redescription R that renames internal symbols while preserving $\sim$.

Proof. Because admissible redesciptions preserve standing, preserve typing envelope, and permit alpha-equivalence only, any bijective renaming of internal representation symbols that preserves well-typedness and introduces/removes no anchors is admissible. If evaluation depends on raw labels not quotiented by alpha-equivalence, applying such a renaming changes the evaluation result while preserving identity equivalence up to alpha-equivalence. Hence such an admissible R exists. $\square$

Lemma 3 (Mediator Necessity). Let S be redescription-stable. Then transition semantics factor through some mediator P .

Proof. Assume no mediator exists. Then there exists $T \in \mathcal{E}(S)$ such that the evaluation of $T(S)$ depends on internal representation labels beyond alpha-equivalence (otherwise one can factor through the quotient-by-alpha-equivalence summary of the required label-invariant information). By the Nontrivial Admissible Renaming Lemma, there exists an admissible redescription R that renames internal labels while preserving $\sim$. Because T is label-sensitive beyond alpha-equivalence, the two computations differ beyond alpha-equivalence, so

$$R(T(S)) \not\sim T(R(S)).$$

This contradicts redescription stability. Therefore a mediator P must exist. $\square$

Definition 7 (Invariant Reference Structure). A reference structure I is *invariant* under admissible redesciptions if for every admissible R ,

$$R(I) \sim I.$$

Lemma 4 (Invariant Reference Necessity). Let S be redescription-stable and let P be a mediator for S . Then there exists an invariant reference structure I .

Proof. Assume no invariant reference structure exists. Then there exists an admissible redescription R such that any candidate reference resolution differs under R while preserving alpha-equivalence at the raw representation level.

Because transitions factor through the mediator P , consistency of transition composition and identity persistence must be computed relative to a redescription-invariant reference resolution; otherwise the induced transition operators $\hat{T}$ cannot be well-defined on identity-equivalence classes under redescription. Concretely, under reference drift there exists $T \in \mathcal{E}(S)$ whose induced evaluation (via P) depends on the choice of reference resolution, yielding

$$R(T(S)) \not\sim T(R(S)).$$

This contradicts redescription stability. Therefore an invariant reference structure I must exist. $\square$

3.4 Single-System Projection Necessity and Necessity-Under-Removal

Theorem 1 (Single-System Projection Necessity). Let S be redescription-stable. Then there exist structures (P, I) such that:

- (i) for every $T \in \mathcal{E}(S)$, $T(S) = \hat{T}(P(S))$ for some induced $\hat{T}$,
- (ii) for every admissible R , $R(I) \sim I$,
- (iii) identity equivalence is preserved under transition composition (by redescription stability).

Proof. By Lemma (Mediator Necessity), there exists a mediator P . By Lemma (Invariant Reference Necessity), there exists an invariant reference structure I . Redescription stability provides commutation for all $T \in \mathcal{E}(S)$ and admissible R , hence preserves identity equivalence under compositions of well-typed transitions. $\square$

Theorem 2 (Necessity Under Removal (Single-System)). If either (a) mediator factorization is absent or (b) invariant reference is absent, then there exist admissible R and $T \in \mathcal{E}(S)$ such that redescription stability fails:

$$R(T(S)) \not\sim T(R(S)).$$

Proof. (a) Without mediator factorization, some transition evaluation remains label-sensitive beyond alpha-equivalence, so the Nontrivial Admissible Renaming Lemma yields an admissible R producing commutation failure for that transition.

(b) Without invariant reference, reference resolution can drift under admissible redescrptions; by the Invariant Reference Necessity proof, there exists T for which induced evaluation differs beyond alpha-equivalence, producing commutation failure. $\square$

4 Role-Based Classification (Non-Primitive Naming)

Definition 8 (Agency-Structure Predicate). For a projection structure (P, I) , define $A(P, I)$ to hold if irreversible transitions are evaluated relative to invariant reference I .

Definition 9 (Intentional-Structure Predicate). Define $M(P, I)$ to hold if transition evaluation includes state-target resolution mediated through P .

Definition 10 (Interior-Representation Predicate). Define $Q(P, I)$ to hold if mediator P provides compressed coordination sufficient for redescription stability.

Lemma 5 (Role Predicates Hold for Single-System Projection Structures). If (P, I) satisfies the Single-System Projection Necessity theorem, then $A(P, I)$, $M(P, I)$, and $Q(P, I)$ hold.

Proof. Factorization through P implies transition evaluation is mediated (hence supports state-target resolution via the induced operators), and invariance of I implies evaluation is relative to a redescription-invariant reference. The existence of a mediator that blocks label-sensitivity beyond alpha-equivalence supplies the required compression for stable coordination. Thus A , M , and Q hold. $\square$

Terminology Convention. We refer to A as *agency-structure*, M as *intentional-structure*, and Q as *interior-representation-structure*. These are role-names only and introduce no new primitives.

5 Illustrative Example (Non-Primitive)

This section is illustrative only and introduces no new primitives or assumptions.

5.1 Example: Neural State Transition as Mediated Evaluation

Consider a biological cognitive system whose internal state transitions are described at some representational level as neural state updates.

Let S denote the current neural state and let $T \in \mathcal{E}(S)$ represent a well-typed neural transition (e.g., synaptic update under lawful dynamics).

If transition evaluation depended directly on raw neuron labels rather than label-invariant structure, then an admissible redescription R (renaming neuron indices without altering functional organization) would yield:

$$R(T(S)) \not\sim T(R(S)),$$

violating redescription stability.

Therefore transition evaluation must factor through a mediator P that captures only the label-invariant structural content required for lawful update.

The invariant reference I corresponds structurally to a stable index under which successive neural transitions are attributed consistently across admissible redesigns.

This example does not introduce neuroscience as a primitive. It merely instantiates the formal structure in a familiar domain.

6 Multi-System Interaction (Fully Formal)

6.1 Joint Typing, Equivalence, and Stability

Definition 11 (Joint System). Let S_1, S_2 be systems. The joint system is

$$S_{12} := (S_1, S_2).$$

Definition 12 (Joint Envelope). Define $\mathcal{E}_{12}(S_{12})$ to be the set of joint transition operators J such that

$$J(S_{12}) = (S'_1, S'_2)$$

is well-defined and yields a joint system of the same arity. This is a typing/applicability domain only.

Definition 13 (Joint Identity Equivalence). For joint systems $S_{12} = (S_1, S_2)$ and $S'_{12} = (S'_1, S'_2)$, write

$$S_{12} \sim S'_{12}$$

iff:

- (i) $S_1 \sim S'_1$ and $S_2 \sim S'_2$,
- (ii) standing matches componentwise,
- (iii) $\mathcal{E}_{12}(S_{12}) = \mathcal{E}_{12}(S'_{12})$,
- (iv) for every $J \in \mathcal{E}_{12}(S_{12})$, the outcomes $J(S_{12})$ and $J(S'_{12})$ agree up to alpha-equivalence only.

No semantic drift beyond alpha-equivalence is permitted.

Definition 14 (Admissible Joint Redescription). A transformation R_{12} on joint representations is admissible if:

- (i) standing preserved componentwise,
- (ii) joint envelope preserved:

$$\mathcal{E}_{12}(S_{12}) = \mathcal{E}_{12}(R_{12}(S_{12})),$$

- (iii) no anchors introduced or removed,
- (iv) joint identity equivalence under $\sim$ preserved (alpha-equivalence only).

Definition 15 (Joint Redescription Stability). S_{12} is jointly redescription-stable if for all admissible R_{12} and all $J \in \mathcal{E}_{12}(S_{12})$,

$$R_{12}(J(S_{12})) \sim J(R_{12}(S_{12})).$$

6.2 Componentwise Closure and Well-Formedness

Lemma 6 (Componentwise Closure of Joint Admissibility). Let R_1 be an admissible redescription on S_1 and R_2 an admissible redescription on S_2 . Define

$$R_{12}(S_{12}) := (R_1(S_1), R_2(S_2)).$$

Then R_{12} is an admissible joint redescription.

Proof. Standing preservation holds componentwise. Typing preservation of each component implies typing preservation on the ordered pair, hence the joint envelope is preserved. No anchors are introduced/removed componentwise, hence none jointly. Alpha-equivalence-only identity preservation holds componentwise and therefore jointly. Thus R_{12} satisfies the admissibility conditions. $\square$

Lemma 7 (Well-Formedness of Joint Stability). Let R_{12} be admissible and $J \in \mathcal{E}_{12}(S_{12})$. Then $R_{12}(J(S_{12}))$ and $J(R_{12}(S_{12}))$ are well-defined, and the stability statement is well-formed.

Proof. Since $J \in \mathcal{E}_{12}(S_{12})$, $J(S_{12})$ is well-defined. Admissible R_{12} preserves typing, hence $R_{12}(J(S_{12}))$ is well-defined. Envelope preservation implies $J \in \mathcal{E}_{12}(R_{12}(S_{12}))$, hence $J(R_{12}(S_{12}))$ is well-defined. Therefore the relation is well-formed. $\square$

6.3 Joint Irreversibility and Nontrivial Renaming

Definition 16 (Joint Irreversible Transition). A joint transition $J \in \mathcal{E}_{12}(S_{12})$ is irreversible if there does not exist admissible $K \in \mathcal{E}_{12}(J(S_{12}))$ such that

$$K(J(S_{12})) \sim S_{12}.$$

Lemma 8 (Nontrivial Admissible Joint Renaming). If evaluation of some $J \in \mathcal{E}_{12}(S_{12})$ depends on internal labels beyond alpha-equivalence, then there exists an admissible joint redescription R_{12} that renames internal symbols while preserving $\sim$.

Proof. As in the single-system case: any bijective renaming preserving well-typedness, standing, and anchors is admissible. If evaluation depends on raw labels beyond alpha-equivalence, such a renaming changes evaluation while preserving $\sim$. $\square$

6.4 Joint Mediators and Joint Invariant References

Definition 17 (Joint Transition Mediator). A structure P_{12} is a joint mediator if for every $J \in \mathcal{E}_{12}(S_{12})$ there exists induced $\hat{J}$ such that

$$J(S_{12}) = \hat{J}(P_{12}(S_{12})).$$

Lemma 9 (Joint Mediator Necessity). Let S_{12} be jointly redescription-stable. Then there exists a joint mediator P_{12} .

Proof. Assume no joint mediator exists. Then some $J \in \mathcal{E}_{12}(S_{12})$ is label-sensitive beyond alpha-equivalence. By the Nontrivial Admissible Joint Renaming Lemma, there exists admissible R_{12} renaming internal labels while preserving $\sim$. Because J is label-sensitive, the two computations differ beyond alpha-equivalence, yielding

$$R_{12}(J(S_{12})) \not\sim J(R_{12}(S_{12})),$$

contradicting joint redescription stability. Therefore a joint mediator exists. $\square$

Definition 18 (Joint Invariant Reference Pair). A pair (I_1, I_2) is jointly invariant if for every admissible joint redescription R_{12} ,

$$R_{12}(I_1, I_2) \sim (I_1, I_2).$$

Lemma 10 (Joint Invariant Reference Necessity). Let S_{12} be jointly redescription-stable and let P_{12} be a joint mediator. Then there exists a jointly invariant reference pair (I_1, I_2) .

Proof. Assume no jointly invariant reference pair exists. Then there exists admissible R_{12} such that reference resolution drifts while preserving alpha-equivalence at the representation level:

$$R_{12}(I_1, I_2) \not\sim (I_1, I_2).$$

Because joint transitions factor through P_{12} , stable evaluation under composition requires jointly invariant reference resolution; otherwise the induced operators cannot be well-defined on joint identity equivalence classes under admissible redescriptions. Therefore there exists $J \in \mathcal{E}_{12}(S_{12})$ such that induced evaluation differs beyond alpha-equivalence, yielding

$$R_{12}(J(S_{12})) \not\sim J(R_{12}(S_{12})),$$

contradicting joint stability. Hence a jointly invariant reference pair exists. $\square$

6.5 Joint Projection Necessity and Necessity-Under-Removal

Theorem 3 (Interaction Projection Necessity (Two-System)). Let S_{12} be jointly redescription-stable. Then:

- (i) each component S_i is redescription-stable as a single system under admissible componentwise redescriptions, and thus instantiates (P_i, I_i) ,
- (ii) there exists a joint mediator P_{12} ,
- (iii) there exists a jointly invariant reference pair (I_1, I_2) .

Proof. (i) Let R_1 be an admissible redescription on S_1 and let R_2 be identity on S_2 . By the Componentwise Closure Lemma, the induced R_{12} is admissible. Consider a joint transition J that acts as some $T \in \mathcal{E}(S_1)$ on the first component and as identity on the second. Such J is well-typed and in $\mathcal{E}_{12}(S_{12})$. Joint stability gives

$$R_{12}(J(S_{12})) \sim J(R_{12}(S_{12})).$$

Projecting to the first component yields $R_1(T(S_1)) \sim T(R_1(S_1))$ for all admissible R_1 and all $T \in \mathcal{E}(S_1)$, i.e. S_1 is redescription-stable. Similarly for S_2 . By the single-system theorems, each component instantiates (P_i, I_i) .

(ii) Joint mediator existence is Lemma (Joint Mediator Necessity).

(iii) Joint invariant reference pair existence is Lemma (Joint Invariant Reference Necessity). $\square$

Theorem 4 (Necessity Under Removal (Joint)). If either (a) joint mediator factorization is absent or (b) joint invariant reference fails, then there exist admissible R_{12} and $J \in \mathcal{E}_{12}(S_{12})$ such that joint redescription stability fails:

$$R_{12}(J(S_{12})) \not\sim J(R_{12}(S_{12})).$$

Proof. (a) Without mediator factorization, some joint transition evaluation remains label-sensitive beyond alpha-equivalence, so the Nontrivial Admissible Joint Renaming Lemma yields an admissible R_{12} producing commutation failure.

(b) Without joint invariant reference, reference resolution can drift under admissible joint redescriptions; by the Joint Invariant Reference Necessity proof, there exists J producing commutation failure. $\square$

7 Extension to n Systems (Fully Formal)

7.1 Restriction Maps and Restricted Actions

Definition 19 (Pairwise Restriction and Inclusion). Let $S_{1\dots n} = (S_1, \dots, S_n)$ be a joint system. For indices $i \neq j$, define

$$\pi_{ij}(S_{1\dots n}) := (S_i, S_j) =: S_{ij}.$$

Define the inclusion map ι_{ij} from S_{ij} into $S_{1\dots n}$ by holding all components other than i, j fixed.

Definition 20 (Restricted Redescription). Let $R_{1\dots n}$ be a joint redescription on $S_{1\dots n}$. Define its restriction to (i, j) by

$$R_{ij} := \pi_{ij} \circ R_{1\dots n} \circ \iota_{ij},$$

whenever well-typed.

Definition 21 (Restricted Joint Transition). Let $J_{1\dots n}$ be a joint transition on $S_{1\dots n}$. If $J_{1\dots n}$ acts nontrivially only on components (i, j) and as identity on all others, define its restriction J_{ij} by

$$J_{ij}(S_{ij}) := \pi_{ij}(J_{1\dots n}(\iota_{ij}(S_{ij}))).$$

7.2 Inherited Pairwise Stability

Lemma 11 (Inherited Pairwise Stability). Assume $S_{1\dots n}$ is jointly redescription-stable (in the natural n -system extension). Fix $i \neq j$. Then the pairwise subsystem S_{ij} is jointly redescription-stable under admissible restricted redescrptions and well-typed restricted transitions:

$$R_{ij}(J_{ij}(S_{ij})) \sim J_{ij}(R_{ij}(S_{ij})).$$

Proof. Let R_{ij} be any admissible redescription on S_{ij} . Extend it to an admissible $R_{1\dots n}$ that acts as R_{ij} on components (i, j) and as identity on all others. Let J_{ij} be any well-typed joint transition on S_{ij} and extend it to $J_{1\dots n}$ acting as J_{ij} on (i, j) and as identity on all other components. Apply joint stability of $S_{1\dots n}$ to $R_{1\dots n}$ and $J_{1\dots n}$, then restrict both sides by π_{ij} . Typing preservation and alpha-equivalence-only equivalence ensure the restricted commutation holds. $\square$

Corollary 1 (Forced Pairwise Interaction Structures). If $S_{1\dots n}$ is jointly redescription-stable, then for every pair (i, j) the subsystem S_{ij} instantiates a joint mediator P_{ij} and jointly invariant references (I_i, I_j) .

8 Constructive Global Mediation and Compatibility

8.1 Product Mediator Construction

Assume $S_{1\dots n}$ is jointly redescription-stable. By the previous corollary, for each pair (i, j) a joint mediator P_{ij} exists.

Definition 22 (Product Mediator). Define the product mediator $P_{1\dots n}^\Pi$ by

$$P_{1\dots n}^\Pi(S_{1\dots n}) := (P_{ij}(\pi_{ij}(S_{1\dots n})))_{1 \leq i < j \leq n}.$$

8.2 Triple Compatibility

Definition 23 (Triple Compatibility (Consistency Constraint)). The family $\{P_{ij}\}$ is triple-compatible if for all distinct i, j, k and all states $S_{1\dots n}$, the induced pairwise mediator outputs agree on overlaps up to $\sim$, i.e., whenever well-typed,

$$P_{ij}(\pi_{ij}(S_{1\dots n})) \sim \pi_{ij}(P_{ik}(\pi_{ik}(S_{1\dots n}))),$$

and similarly with (ik) replaced by (jk) .

8.3 Pairwise Determinacy Forced by Stability

Definition 24 (Pairwise Indistinguishability). For joint states X, Y of $S_{1\dots n}$, write

$$X \equiv_\Pi Y$$

iff their product mediator values agree up to $\sim$, i.e.

$$P_{1\dots n}^\Pi(X) \sim P_{1\dots n}^\Pi(Y),$$

meaning each pairwise component satisfies

$$P_{ij}(\pi_{ij}(X)) \sim P_{ij}(\pi_{ij}(Y)) \quad \text{for all } i < j.$$

Definition 25 (Pairwise-Preserving Renaming). An admissible joint redescription $R_{1\dots n}$ is a *pairwise-preserving renaming* if:

- (i) $R_{1\dots n}$ is an admissible joint redescription,
- (ii) $R_{1\dots n}$ acts as a pure alpha-renaming on internal representation symbols (no semantic drift),
- (iii) for all joint states X ,

$$P_{1\dots n}^\Pi(R_{1\dots n}(X)) \sim P_{1\dots n}^\Pi(X).$$

Lemma 12 (Renaming Invariance of Identity Equivalence). If R is an admissible redescription that is a pure alpha-renaming, then for every well-typed state Z ,

$$R(Z) \sim Z.$$

Proof. By definition, a pure alpha-renaming preserves standing, preserves typing, introduces/removes no anchors, and changes only symbol names. Identity equivalence $\sim$ permits alpha-equivalence only, and forbids semantic drift beyond alpha-equivalence. Therefore $R(Z) \sim Z$. $\square$

Lemma 13 (Pairwise-Renaming Richness). Assume $\{P_{ij}\}$ is triple-compatible. If $X \equiv_\Pi Y$, then there exists a pairwise-preserving renaming $R_{1\dots n}$ such that

$$R_{1\dots n}(X) \sim Y.$$

Proof. By definition of $X \equiv_\Pi Y$, all pairwise mediator values agree up to $\sim$. Triple compatibility ensures these pairwise agreements are mutually consistent on overlaps.

Because admissible redesciptions include alpha-renamings that preserve typing and anchors, we may choose a renaming that aligns the internal symbol presentation of X with that of Y while leaving each pairwise mediator component invariant up to $\sim$. Such a renaming is admissible, pairwise-preserving by construction, and yields $R_{1\dots n}(X) \sim Y$. $\square$

Lemma 14 (Pairwise Determinacy of Joint Transitions). Assume $S_{1\dots n}$ is jointly redescription-stable and $\{P_{ij}\}$ is triple-compatible. Then for every well-typed joint transition $J_{1\dots n}$ and all joint states X, Y ,

$$X \equiv_\Pi Y \Rightarrow J_{1\dots n}(X) \sim J_{1\dots n}(Y).$$

Proof. Let $X \equiv_\Pi Y$. By the Pairwise-Renaming Richness lemma, there exists a pairwise-preserving renaming $R_{1\dots n}$ such that $R_{1\dots n}(X) \sim Y$. Since $R_{1\dots n}$ is an admissible joint redescription, joint redescription stability gives

$$R_{1\dots n}(J_{1\dots n}(X)) \sim J_{1\dots n}(R_{1\dots n}(X)).$$

Using $R_{1\dots n}(X) \sim Y$ and preservation of $\sim$ under admissible redescription, we have

$$J_{1\dots n}(R_{1\dots n}(X)) \sim J_{1\dots n}(Y).$$

Because $R_{1\dots n}$ is a pure alpha-renaming, Renaming Invariance implies

$$R_{1\dots n}(J_{1\dots n}(X)) \sim J_{1\dots n}(X).$$

By transitivity of $\sim$, we conclude $J_{1\dots n}(X) \sim J_{1\dots n}(Y)$. $\square$

8.4 Global Mediation via Product Mediator

8.5 Syntactic Supply and Envelope Non-Minimality (Strengthened)

Definition 26 (Fresh-Name Supply). Fix a typed representation signature in which, for each syntactic type τ , there exists an infinite set of names $\text{Name}(\tau)$. An alpha-renaming is a capture-avoiding bijection on names that respects types.

Lemma 15 (Alpha-Renaming Admissibility). Let ρ be a typed capture-avoiding alpha-renaming that preserves typing and does not introduce or remove anchors. Then ρ is an admissible redescription (at any arity).

Proof. By construction, ρ preserves standing (it changes no semantic content), preserves typing envelope (pure renaming), introduces/removes no anchors, and preserves identity equivalence under alpha-equivalence-only definition. Therefore ρ satisfies the admissibility conditions. $\square$

Definition 27 (Statewise Envelope Non-Minimality). The joint representation is statewise envelope-non-minimal if for every well-typed state X and a non-identity alpha-renaming ρ such that:

- (i) $\rho(X)$ is well-typed,
- (ii) $\mathcal{E}_{1\dots n}(X) = \mathcal{E}_{1\dots n}(\rho(X))$,
- (iii) $X \sim \rho(X)$.

Definition 28 (Name Support). For a well-typed state X , define $\text{Names}(X)$ to be the finite set of names appearing in its representation. For a mediator value $P_{ij}(\pi_{ij}(X))$, define its name support analogously. The residual name-set of X relative to pairwise mediators is

$$\text{Res}(X) := \text{Names}(X) \setminus \bigcup_{i < j} \text{Names}(P_{ij}(\pi_{ij}(X))).$$

Lemma 16 (Simultaneous Pairwise Alignment). Assume triple compatibility and fresh-name supply. Let X, Y satisfy $X \equiv_{\Pi} Y$. Then there exists a capture-avoiding alpha-renaming ρ such that for all $i < j$,

$$P_{ij}(\pi_{ij}(\rho(X))) \sim P_{ij}(\pi_{ij}(Y)).$$

Proof. For each pair (i, j) , triple compatibility ensures the mediator outputs agree on overlaps. Fresh-name supply guarantees that renamings may be chosen to avoid capture across overlapping name-sets. Proceed inductively over the finite set of pairs, extending the renaming at each step while avoiding previously assigned names. Because overlaps are consistent and the name supply is infinite per type, the extension process terminates without conflict. Thus a single capture-avoiding alpha-renaming simultaneously aligns all pairwise mediator presentations. $\square$

Lemma 17 (Strong Pairwise-Renaming Richness). Assume $\{P_{ij}\}$ is triple-compatible and the representation is statewise envelope-non-minimal. If $X \equiv_{\Pi} Y$, then there exists an admissible pairwise-preserving alpha-renaming $R_{1\dots n}$ such that

$$R_{1\dots n}(X) \sim Y.$$

Proof. Fresh-name supply ensures alpha-renamings may be chosen independently on disjoint name-sets.

Given $X \equiv_{\Pi} Y$, triple compatibility ensures pairwise mediator outputs are consistent on overlaps.

Construct $R_{1\dots n}$ in two stages: (1) align pairwise mediator presentations via alpha-renaming; (2) adjust residual name-sets using fresh names guaranteed by supply. Envelope non-minimality ensures nontrivial alpha-variants exist within the same envelope and equivalence class.

The resulting renaming preserves typing, preserves envelope, introduces no anchors, and preserves identity equivalence up to alpha-equivalence only. Thus $R_{1\dots n}(X) \sim Y$. $\square$

Lemma 18 (Strong Pairwise Determinacy). Assume $S_{1\dots n}$ is jointly redescription-stable, $\{P_{ij}\}$ is triple-compatible, and the representation is statewise envelope-non-minimal. Then for every well-typed joint transition $J_{1\dots n}$ and all joint states X, Y ,

$$X \equiv_{\Pi} Y \Rightarrow J_{1\dots n}(X) \sim J_{1\dots n}(Y).$$

Proof. Let $X \equiv_{\Pi} Y$. By Strong Pairwise-Renaming Richness, there exists an admissible pairwise-preserving renaming $R_{1\dots n}$ such that $R_{1\dots n}(X) \sim Y$.

Joint redescription stability gives

$$R_{1\dots n}(J_{1\dots n}(X)) \sim J_{1\dots n}(R_{1\dots n}(X)).$$

Renaming invariance implies

$$R_{1\dots n}(J_{1\dots n}(X)) \sim J_{1\dots n}(X).$$

Transitivity yields

$$J_{1\dots n}(X) \sim J_{1\dots n}(Y).$$

$\square$

8.6 Universal Global Factorization

Theorem 5 (Universal Global Factorization Under Triple Compatibility). Assume $S_{1\dots n}$ is jointly redescription-stable and $\{P_{ij}\}$ is triple-compatible. Then for every well-typed joint transition $J_{1\dots n}$ there exists an induced operator $\hat{J}_{1\dots n}^{\Pi}$ such that

$$J_{1\dots n}(S_{1\dots n}) \sim \hat{J}_{1\dots n}^{\Pi}(P_{1\dots n}^{\Pi}(S_{1\dots n})).$$

Proof. Define $\hat{J}_{1\dots n}^{\Pi}$ on product-mediator values as follows. Given a product-mediator value v in the image of $P_{1\dots n}^{\Pi}$, choose any joint state X with $P_{1\dots n}^{\Pi}(X) = v$ (well-typedness guarantees existence for values in the image) and set

$$\hat{J}_{1\dots n}^{\Pi}(v) := J_{1\dots n}(X).$$

We must show this is well-defined up to $\sim$. If $P_{1\dots n}^{\Pi}(X) \sim P_{1\dots n}^{\Pi}(Y)$, then $X \equiv_{\Pi} Y$ by definition. By the Pairwise Determinacy lemma, $J_{1\dots n}(X) \sim J_{1\dots n}(Y)$. Therefore $\hat{J}_{1\dots n}^{\Pi}$ is well-defined up to $\sim$, and

$$J_{1\dots n}(S_{1\dots n}) \sim \hat{J}_{1\dots n}^{\Pi}(P_{1\dots n}^{\Pi}(S_{1\dots n})).$$

$\square$

Corollary 2 (Obstruction). If triple compatibility fails for some triple (i, j, k) , then no global mediator exists whose pairwise restrictions agree with all P_{ij} up to $\sim$.

Proof. Failure of triple compatibility yields contradictory pairwise mediator outputs on a shared overlap beyond alpha-equivalence. Any purported global mediator agreeing with each P_{ij} would induce compatible overlap restrictions, contradicting the failure. Therefore such a global mediator cannot exist. $\square$

9 Capstone Closure

Theorem 6 (Multi-Agent Mentis Umbra Closure). Let $S_{1\dots n}$ be jointly redescription-stable. Then:

- (i) each component S_i instantiates a projection structure (P_i, I_i) and satisfies $A(P_i, I_i)$, $M(P_i, I_i)$, and $Q(P_i, I_i)$,
- (ii) each pair (i, j) instantiates a joint mediator structure (P_{ij}, I_i, I_j) ,
- (iii) a constructive global mediator exists via the product mediator $P_{1\dots n}^\Pi$ if triple compatibility holds,
- (iv) failure of triple compatibility yields an obstruction to global coherent mediation.

Proof. (i) follows from single-system stability inherited from joint stability (two-system theorem and pairwise inheritance), together with the role-predicate lemma. (ii) follows from the forced pairwise interaction structures corollary. (iii) follows from the Constructive Global Mediator theorem. (iv) follows from the Obstruction corollary. $\square$

10 Discussion and Structural Implications

10.1 Interpretive Consequences

The results obtained are not explanatory reductions but structural constraints.

Any system satisfying irreversibility and redescription stability must instantiate:

- (i) a transition mediator P ,
- (ii) an invariant reference structure I ,
- (iii) interaction mediators in multi-system contexts,
- (iv) a compatibility condition for global coherence.

These are not optional features. They are enforced by the commutation condition $R(T(S)) \sim T(R(S))$ and its joint-system extensions.

Accordingly:

- What is ordinarily described as agency corresponds structurally to invariant-reference-mediated transition evaluation.
- What is described as intentionality corresponds to mediator-governed state–target resolution.
- What is described as qualitative interiority corresponds to compression sufficient to block label-sensitive divergence under admissible redescription.

No additional ontological commitments are required for these classifications.

10.2 Structural, Not Metaphysical

This work does not attempt to explain phenomenology in experiential terms. It establishes necessary structural conditions for stable evaluation under irreversibility.

The distinction is critical:

The framework constrains what any stable system must instantiate; it does not posit new entities beyond those required for admissible coordination.

10.3 Global Coherence

For multi-agent systems, coherence is not assumed.

It exists if and only if pairwise mediator structures satisfy triple compatibility, allowing construction of a global mediator via the product construction.

Failure of triple compatibility yields formal obstruction, not metaphorical fragmentation.

10.4 Standalone Formal Contribution

Independent of any broader corpus commitments, this paper establishes:

- (i) necessity of mediator factorization under redescription stability,
- (ii) necessity of invariant reference structures,
- (iii) necessity of joint mediation in interacting systems,
- (iv) algebraic compatibility criterion for global mediation.

These results follow solely from the definitions provided herein.

10.5 Interpretive Implications

The preceding results classify mind-level and social-level structures as projection artifacts required by admissibility and redescription stability.

No claim is made that:

- projection structures are reducible to a particular substrate,
- phenomenology is eliminated,
- free will is metaphysically denied,
- social coherence is normatively prescribed.

The results are strictly structural: any system satisfying irreversibility and redescription stability must instantiate projection and mediation structures of the form derived.

10.6 Scope Limitations

This paper does not:

- attempt a phenomenological analysis,
- provide empirical neuroscience,
- construct computational implementations,
- analyze probabilistic or stochastic extensions,
- address semantic content beyond structural role classification.

Future work may investigate probabilistic envelopes, computational realizability constraints, and quantitative stability metrics, but such extensions would operate within the existing admissibility framework.

10.7 Standalone Position

Even independent of broader corpus commitments, the results establish:

- (i) necessity of mediator structures under redescription stability,
- (ii) necessity of invariant reference structures,
- (iii) necessity of interaction mediators in multi-system contexts,
- (iv) a constructive global mediation criterion.

These are formal consequences of the definitions given herein.

Non-Expansion Clause

This paper introduces no new primitives, no new enforcement tools, and no ontic commitments. All results are classification consequences of admissibility, typing-envelope preservation, alpha-equivalence-only identity equivalence, and redescription stability.

A Formal Dependency Graph

This appendix summarizes logical dependencies among definitions, lemmas, and theorems. The graph is acyclic and layered.

Layer 1: Typing and Equivalence Foundations

- Single-System Envelope $\mathcal{E}(S)$
- Irreversible Transition
- Identity Equivalence $\sim$
- Admissible Redescription
- Redescription Stability

These define the commutation condition:

$$R(T(S)) \sim T(R(S)).$$

Layer 2: Single-System Structural Necessity

- Nontrivial Admissible Renaming Lemma
- Mediator Necessity Lemma
- Invariant Reference Necessity Lemma
- Single-System Projection Necessity Theorem
- Necessity Under Removal (Single-System)

Dependencies:

- Mediator Necessity depends on Renaming + Stability.
- Invariant Reference Necessity depends on Mediator + Stability.
- Projection Necessity depends on both necessity lemmas.

Layer 3: Role Classification

- Agency-Structure Predicate A
- Intentional-Structure Predicate M
- Interior-Representation Predicate Q

These depend solely on (P, I) and introduce no new primitives.

Layer 4: Two-System Interaction

- Joint Envelope $\mathcal{E}_{12}$
- Joint Identity Equivalence
- Admissible Joint Redescription
- Joint Redescription Stability
- Joint Mediator Necessity
- Joint Invariant Reference Necessity
- Interaction Projection Necessity
- Necessity Under Removal (Joint)

Dependencies:

- Joint Mediator Necessity depends on Joint Renaming + Joint Stability.
- Joint Invariant Reference depends on Joint Mediator + Joint Stability.
- Interaction Projection Necessity depends on Single-System results + Joint Necessities.

Layer 5: n -System Extension

- Pairwise Restriction Maps
- Inherited Pairwise Stability Lemma
- Forced Pairwise Interaction Structures Corollary

Dependencies:

- Inherited Stability depends on Joint Stability.
- Pairwise structures depend on Two-System Necessity.

Layer 6: Global Mediation

- Product Mediator Definition
- Triple Compatibility Definition
- Constructive Global Mediator Theorem
- Obstruction Corollary

Dependencies:

- Product Mediator depends on pairwise mediators.
- Global Mediation depends on Triple Compatibility + Pairwise Necessity.

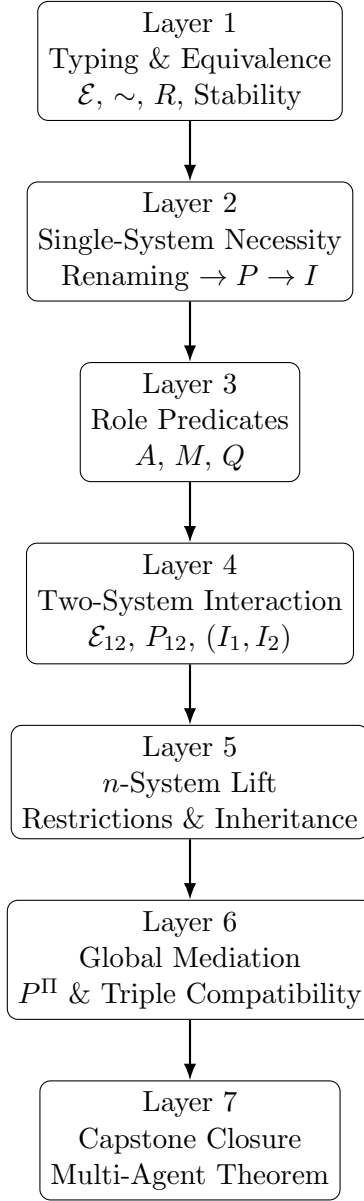
Layer 7: Capstone Closure

- Multi-Agent Mentis Umbra Closure Theorem

Dependencies:

- Closure depends on all prior layers.

Compact Dependency Diagram



Graph Summary

Logical flow proceeds strictly upward:

Typing $\rightarrow$ Stability $\rightarrow$ Mediator/Reference $\rightarrow$ Role Classification $\rightarrow$ Joint Mediation $\rightarrow$ Pairwise
Inheritance $\rightarrow$ Global Compatibility $\rightarrow$ Closure

No circular dependencies occur.

A Core Framework Axioms

A.1 Identity. $\sim$ permits alpha-equivalence only.

A.2 Admissible Redescription. Redescription preserves typing, anchors, role assignments, and compatibility.

A.3 Redescription Stability.

$$R(T(S)) \sim T(R(S)).$$

A.4 Standing. Standing is binary, scope-global, conserved, and absorbing.

A.5 Non-Triviality. Admissibility must attach to conserved structural consequences.

A.6 System Boundary Rule. All data consulted by a mechanism are part of the system state and must be acted upon by admissible redescrptions.

Quotient Minimality and Conditional Reference Necessity

Quotient Universal Property. Any redescription-stable evaluation $f : S \rightarrow Y$ constant on $\approx$ -classes uniquely factors through the quotient $\pi : S \rightarrow S/\approx$. The quotient is therefore the minimal mediator: any alternative mediator factors through it.

Conditional Reference Necessity. If a system assigns *single-valued representatives* in S to quotient states (rather than working purely in $S/\approx$), then an admissibly invariant section $\kappa : S/\approx \rightarrow S$ is required to preserve redescription stability under composition. Absent this condition, representative choice need not be unique and no invariant resolver is forced.

Irreversibility Scope Note. Irreversibility is not required for descent of transitions to the quotient (which follows from stability alone). It is used only to exclude trivial "repair by backtracking" when discussing single-valued representative selection and persistence under composition.